

The **upper airway** is the main plausible deposition site; a precise **upper-versus-lower fraction for inhaled norovirus** has not been measured.

1. Introduction

Norovirus aerosol transmission is biologically plausible, but the pathway the literature supports is indirect: aerosolized virus from vomiting, toilet flushing, contaminated floors, or wastewater is inhaled, deposits mainly in the extrathoracic and conducting airways depending on particle size, and then can be swallowed in mucus and reach the intestine, where norovirus replication is established (Alsved et al., 2019; Bonifait et al., 2015; Nazaroff, 2011; Xiao et al., 2017; Uhrbrand et al., 2018; Rushton et al., 2019; Green et al., 2020). The literature does not contain a direct human measurement of the fraction of inhaled norovirus that deposits in the upper airway and is subsequently swallowed versus the fraction that deposits in the lower airway and fails to establish infection; that missing quantity is repeatedly visible as an inference gap between aerosol detection studies and gastrointestinal pathogenesis studies (Alsved et al., 2019; Boles et al., 2021; Adiyani et al., 2025; Overbey et al., 2021).

What is known comes from three evidence streams that only partially connect: outbreak and air-sampling studies showing norovirus-containing aerosols exist after vomiting and toilet flushing (Alsved et al., 2019; Boles et al., 2021; Bonifait et al., 2015); aerosol physics showing that deposition site is governed mainly by particle aerodynamic diameter and breathing route (Wang et al., 2021; Thomas, 2013); and norovirus pathogenesis studies showing productive infection in the small intestine, not respiratory epithelium (Marks et al., 2000; Booth & Frost, 2019; Green et al., 2020). Within that synthesis, the best-supported answer is that **infection begins in the gut**, while inhalation serves only as a delivery route to the gut when particles deposit in the upper airway or conducting airways and are then swallowed (Alsved et al., 2019; Booth & Frost, 2019; Xiao et al., 2017; Lockhart et al., 2022).

Does inhaled norovirus mainly reach the intestine by depositing in the upper airway and then being swallowed?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus on upper-airway swallowing pathway for norovirus

2. Methods

The search ran over more than 220 million research papers indexed in Consensus, drawing from Semantic Scholar, PubMed, and other sources. The workflow identified 4,354,438 broad candidate records from the initial query and related query expansions, discovered additional linked papers through citation crawling, screened 239 papers after machine relevance filtering, judged 142 as eligible after deduplication and relevance review, and included the top 100 most relevant papers for synthesis.

The review prioritized studies that directly addressed aerosolized norovirus generation, air sampling, particle size, deposition in the respiratory tract, or gastrointestinal initiation of infection. Because the user asked for the deposition-and-swallowing pathway rather than respiratory clearance alone, the synthesis gave greatest weight to papers connecting aerosol particle size to extrathoracic or conducting-airway deposition, and to papers on norovirus intestinal tropism. Respiratory-virus papers were included only when they provided generalizable deposition physics rather than claims about norovirus biology.

Search Strategy

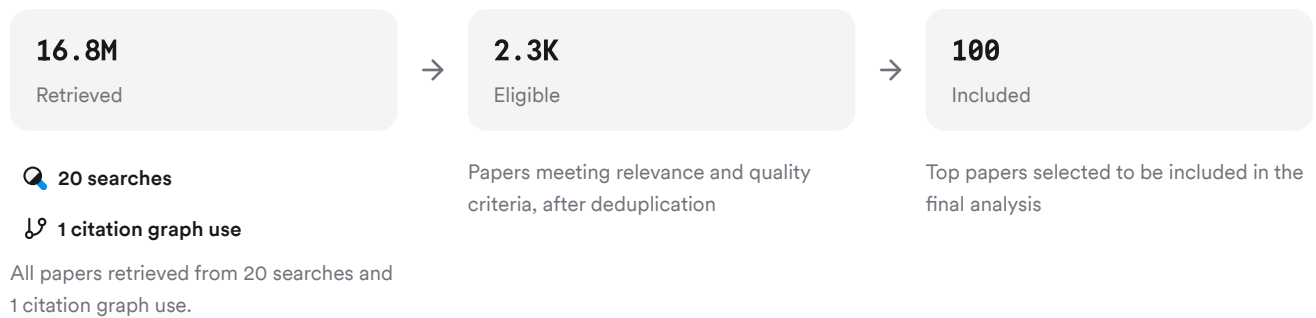


FIGURE 2 Search and screening flow for included papers

Six targeted searches linked norovirus aerosol sources, deposition physics, swallowing-mediated transfer, and intestinal tropism across adjacent literatures.

3. Results

3.1 Key Papers

Three papers anchor this question because together they cover the source aerosol, the inferred deposition route, and the target tissue where infection actually establishes. The strongest direct norovirus paper is Alsved et al., which measured airborne norovirus RNA, size-separated particles, and explicitly linked submicron particles to deposition in the mouth, nose, or conducting airways with likely transport to the gastrointestinal tract (Alsved et al., 2019). Bonifait et al. supplied the field's clearest early articulation that inhaled norovirus would settle in the pharynx and later be swallowed (Bonifait et al., 2015). Green et al. then established that active norovirus replication in humans is found in the small intestine, especially jejunum and ileum, including enteroendocrine epithelial cells (Green et al., 2020).

Paper	Summary
(Alsved et al., 2019)	Submicron particles detected; 10–30% deposit in mouth, nose, or conducting airways (Alsved et al., 2019)
(Bonifait et al., 2015)	Inhaled norovirus can settle in pharynx and be swallowed (Bonifait et al., 2015)
(Green et al., 2020)	Human replication detected in jejunum and ileum epithelial cells (Green et al., 2020)

FIGURE 3 Foundational papers on aerosol, deposition, and tropism

3.2 Source Aerosols

Vomiting is the most consistently supported source of airborne norovirus, with hospital outbreak sampling showing that positive air samples cluster close to recent emesis and that the odds of air positivity rise when sampling occurs within three hours of vomiting (Alsved et al., 2019; Rupprom et al., 2024). Simulated vomiting studies show that virus surrogates can aerosolize during emesis and that the aerosolized fraction can span roughly 7.2×10^{-5} to 2.67×10^{-2} of total expelled virus, depending on viscosity and pressure, which is enough to matter because norovirus infectious dose is low (Tung-Thompson et al., 2015; Boles et al., 2021).

Toilet flushing is also a credible source, especially for fine particles. Murine norovirus aerosols have been collected after flushing at 383–684 RNA copies/m³, and most measured particles from flushing were smaller than 1 µm, with 0.3 µm the most common size (Boles et al., 2021; Tan et al., 2024). Public restroom measurements in non-norovirus studies align with this size range, showing abundant 0.3–3 µm flush aerosols that can rise to breathing height (Schreck et al., 2021).

Air sampling studies in hospitals and wastewater settings repeatedly detect airborne norovirus RNA across a broad concentration range, from about 5 copies/m³ to several thousand copies/m³, confirming that inhalable exposure occurs even though infectivity is rarely measured directly in field settings (Alsved et al., 2019; Bonifait et al., 2015; Kittigul et al., 2025; Stobnicka-Kupiec et al., 2021).

3.3 Deposition Pathway

The deposition route depends mainly on particle size, not on norovirus specifically. Across respiratory deposition models, particles larger than about 5 µm deposit primarily in the nasopharyngeal or oropharyngeal region, often at 87–95%, and are therefore the clearest candidates for the swallow-to-gut pathway (Wang et al., 2021; Dhand & Li, 2020). The question becomes harder for the smaller particles actually observed in norovirus aerosol studies, because vomiting- and flush-generated particles often include a dominant submicron mode (Alsved et al., 2019; Tan et al., 2024; Mori et al., 2017).

For that submicron range, the only norovirus-specific estimate in this corpus is the repeated statement from Alsved and later reviews that particles smaller than 1 µm have a **10–30% probability of deposition in the mouth, nose, or trachea**, from where they are likely to be swallowed and transported to the gastrointestinal tract (Tan et al., 2024; Alsved et al., 2019). That estimate does **not** partition deposition into upper versus lower airway fractions, and it leaves 70–90% either exhaled or deposited elsewhere in the respiratory tract (Alsved et al., 2019; Dhand & Li, 2020).

A useful approximation from general aerosol physics is that 1 µm particles still favor the upper over lower airway, but only modestly. One CFD study reported upper-to-lower deposition ratios of about 4:1 for 1 µm droplets and 15:1 for 10 µm droplets, corresponding to 9.5% upper versus 2.4% lower for 1 µm and 76.7% upper versus 5.1% lower for 10 µm (Shang et al., 2021). Other anatomical simulations likewise found high upper-airway retention for 1–2 µm aerosols under nasal breathing, often exceeding 85% of deposited particles, while higher flow rates shift some deposition toward the pharynx, larynx, and trachea (Shim et al., 2023).

Dimension	>5 µm particles	~1 µm particles	<1 µm particles	Citations
Dominant deposition region	Nasopharynx/oropharynx	Mixed, upper favored	Mixed, limited deposition	(Wang et al., 2021; Dhand & Li, 2020)

Dimension	>5 μm particles	~1 μm particles	<1 μm particles	Citations
Best-supported swallow pathway	Strong	Moderate	Plausible but uncertain	(Nazaroff, 2011; Xiao et al., 2017)
Evidence specific to norovirus	Indirect	Very limited	One 10–30% statement	(Alsved et al., 2019; Tan et al., 2024)
Lower-airway deposition relevance	Limited	Non-zero	Non-zero	(Wang et al., 2021; Dhand & Li, 2020)
Quantified upper/lower split for norovirus	No	No	No	(Alsved et al., 2019; Boles et al., 2021)

FIGURE 4 Particle size determines probable deposition region

Key difference: the literature supports the **direction** of the pathway far better than the **fraction** of dose following it.

3.4 Initiation of Infection

Norovirus infection begins in gastrointestinal tissue, especially the small intestine, not in respiratory epithelium (Marks et al., 2000; Booth & Frost, 2019; Green et al., 2020; Chio et al., 2024). Human tissue data show abundant antigen and RNA in jejunal and ileal tissue, with negative-sense RNA indicating active replication predominantly in intestinal epithelial cells, including chromogranin A–positive enteroendocrine cells (Green et al., 2020). Reviews of enteric immunity also locate host response primarily at the intestinal mucosa (Lockhart et al., 2022).

That means lower-airway deposition is, for this specific question, largely a dead end for primary infection unless particles are cleared upward and swallowed before inactivation. The literature supports mucociliary transport as the mechanism by which inhaled norovirus trapped in upper or conducting airways can be moved toward the larynx and swallowed (Booth & Frost, 2019; Bing-Yuan et al., 2017). Several reviews therefore describe airborne norovirus as a form of **inhalation-mediated ingestion**, not a respiratory infection in the ordinary sense (Hall, 2012; Uhrbrand et al., 2018; Kittigul et al., 2025).

Results Timeline

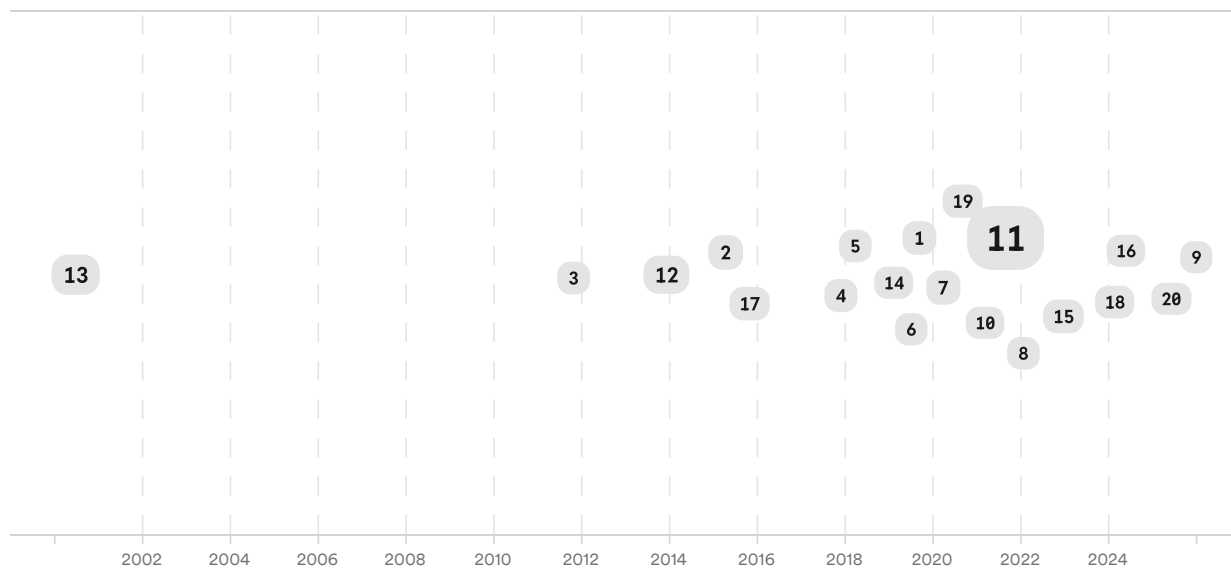


FIGURE 5 Timeline of included studies; larger markers indicate citations

Top Contributors

Type	Name	Papers
Author	Malin Alsved	[b8818e715c0b50d6928ca4983ba47219][d4c21bb7f7895ef0b67fab88c2ba77e3]
Author	Corey Boles	[c749bbb23c77589284b8e553ca519e67][8451d730f58b54d6a832c00fb168eb43]
Author	Yuguo Li	[6f9829f2511f55abb1172fee4cfd1f73][ef23b3eb72725babbfd3f8880bb5fba9]
Journal	<i>Viruses</i>	[e53e94c133a45da69c1d1a7da909dfe2][baad3818a7b45139b9e1060bbec26661] [88bde592e00751bb8fdfe239508c5963][a87210dc17df51a496927ab4176a7b2d]
Journal	<i>Scientific Reports</i>	[d4c21bb7f7895ef0b67fab88c2ba77e3][c749bbb23c77589284b8e553ca519e67] [c03e0b888e915e3cb4147c286a8bd45d][9374e36ec73556d481d0b85fca3c1535]
Journal	<i>Clinical Infectious Diseases</i>	[b8818e715c0b50d6928ca4983ba47219][2f0af7c1b73951edb7f7677df1bb6679]

FIGURE 6 Authors and journals appearing most often

4. Discussion

The central conclusion is fairly robust even though the requested fraction is not known: inhaled norovirus most plausibly establishes infection only after deposition in the upper or conducting airway and subsequent swallowing, with intestinal replication as the biologically relevant endpoint (Alsved et al., 2019; Bonifait et al., 2015; Nazaroff, 2011; Green et al., 2020). Multiple papers independently state this pathway, and none in the corpus provide evidence for productive respiratory epithelial infection by norovirus (Booth & Frost, 2019; Xiao et al., 2017; Marks et al., 2000).

The weakness is quantitative. The corpus measures aerosol generation, environmental concentrations, and sometimes size classes, but almost never couples those measurements to a full dosimetry model that outputs **fraction swallowed to intestine** versus **fraction lost to lower-airway deposition or exhalation** (Alsved et al., 2019; Boles et al., 2021; Adiyani et al., 2025; Overbey et al., 2021). Even the frequently cited 10–30% value applies only to deposition in the mouth, nose, or trachea for submicron particles and should not be misread as “10–30% reaches the intestine” because some deposited material can be exhaled later, inactivated, or remain regionally trapped (Alsved et al., 2019; Tan et al., 2024; Dhand & Li, 2020).

A further complication is that real norovirus aerosols are polydisperse and source-dependent. Vomiting can generate a wide size spectrum including droplets large enough to impact the upper airway strongly, whereas toilet flushing often emphasizes submicron particles, which are more likely to bypass the nose and deposit less efficiently overall (Tung-Thompson et al., 2015; Tan et al., 2024; Schreck et al., 2021). Breathing route also matters: nasal breathing generally increases upper-airway capture relative to oral breathing (Shim et al., 2023; Wedel et al., 2021).

Claim	Evidence Strength	Reasoning	Papers
Norovirus aerosol exposure can plausibly cause GI infection through inhalation followed by swallowing	 Strong	Repeatedly stated across outbreak, review, and air-sampling papers, with no competing mechanism supported	(Alsved et al., 2019; Bonifait et al., 2015; Nazaroff, 2011; Xiao et al., 2017; Uhrbrand et al., 2018)
Infection begins in the small intestine, not respiratory epithelium	 Strong	Human tissue tropism data directly show intestinal replication; respiratory replication is explicitly unsupported in this corpus	(Marks et al., 2000; Green et al., 2020; Lockhart et al., 2022)
Vomiting is the strongest documented aerosol source in outbreaks	 Moderate	Supported by outbreak epidemiology and hospital air sampling, but not all outbreaks exclude fomite spread	(Alsved et al., 2019; Tung-Thompson et al., 2015)
Toilet flushing generates inhalable norovirus-containing aerosols, often submicron	 Moderate	Direct experimental and environmental support exists, but human transmission attribution remains indirect	(Boles et al., 2021; Tan et al., 2024; Rupprom et al., 2024)

Claim	Evidence Strength	Reasoning	Papers
A precise fraction of inhaled norovirus that reaches the intestine after upper-airway deposition is known	Weak	The key quantitative quantity the user asked for has not been directly measured in humans or surrogates	(Alsved et al., 2019; Boles et al., 2021; Overbey et al., 2021)

FIGURE 7 Key claims and evidence across included papers

5. Conclusion

The literature supports a clear qualitative answer: when norovirus is inhaled as an aerosol, the relevant infection pathway is deposition in the upper or conducting airway followed by swallowing to the gut, and productive infection begins in the small intestine rather than the respiratory tract. What it does **not** support is a precise numeric split between “upper-airway deposited and swallowed” and “lower-airway deposited and non-establishing.”

The nearest quantitative statement in the corpus is that **submicron norovirus-containing particles have about a 10–30% chance to deposit in the mouth, nose, or trachea**, where they are likely to be swallowed (Alsved et al., 2019; Tan et al., 2024). That is the best current estimate for the deposition-and-swallowing pathway, but it is not a full intestine-delivery fraction and it does not isolate lower-airway loss.

Research Gaps

The field has characterized aerosol sources and gut tropism better than the dosimetric bridge between them.

	Particle Size	Air Concentration	Regional Deposition	Swallowing Fate	Infectivity
Vomiting	GAP	GAP	GAP	GAP	GAP
Toilet Flush	GAP	GAP	GAP	GAP	GAP
Floor Resuspension	GAP	GAP	GAP	GAP	GAP
Wastewater Aerosol	GAP	GAP	GAP	GAP	GAP
Intestinal Tropism	GAP	GAP	GAP	GAP	GAP

Open Research Questions

The next advances need to connect aerosol physics to enteric inoculation directly rather than inferring the link from separate literatures.

Question	Why
What fraction of inhaled norovirus-containing particles of each aerodynamic size deposits in the nasopharynx, trachea, bronchi, and alveoli during nasal versus oral breathing?	This is the missing dosimetry step needed to answer the user's upper-versus-lower fraction question directly.
Of norovirus deposited in the mouth, nose, and conducting airways, what fraction is transported by mucus and swallowed while still infectious?	Deposition alone does not equal intestinal dose, so infective swallow-through must be measured explicitly.
How do vomiting-generated and toilet-flush-generated size distributions differ in their resulting swallowed intestinal dose?	The two main sources likely produce different deposition patterns, so transmission risk may differ even at similar airborne copy numbers.

In short, inhaled norovirus most likely causes disease by **upper-airway deposition followed by swallowing to the intestine**, but the exact swallowed fraction versus lower-airway loss remains unmeasured.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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